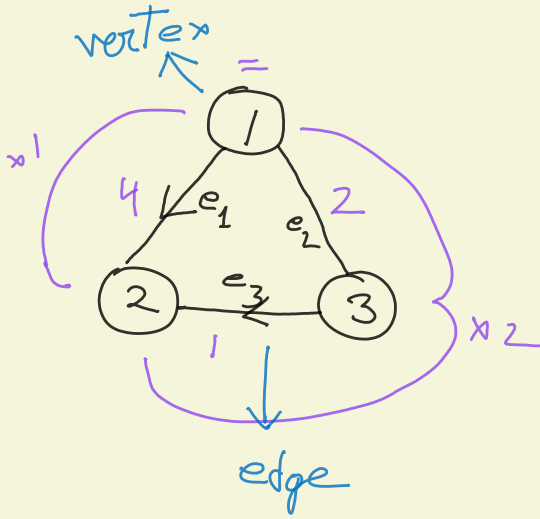


Graph :- A set representation in order is a graph.

$$A = \{3, 6, 9, 10, 11, \dots, n\} \rightarrow \text{set}$$



① with direction ✓

② without direction ✓

③ with weight ✓

$$G = \{ \overset{e_1}{(1, 2)}, \overset{e_2}{(1, 3)}, \overset{e_3}{(3, 2)}, \overset{e_4}{(3, 1)} \}$$

Graph Representation

↳ Matrix $\rightarrow$ loss of memory $a[i][j]$

↳ List (Adjacency List) $\rightarrow$ memory efficient

Matrix Representation

$$G = \{ (1, 2), (2, 3), (1, 4) \}$$

	1	2	3	4
1	0	1	0	3
2	1	0	4	0
3	0	1	0	0
4	1	0	0	0

0 $\rightarrow$ no edge
1 $\rightarrow$ edge present

$$a[1][2] = 1$$

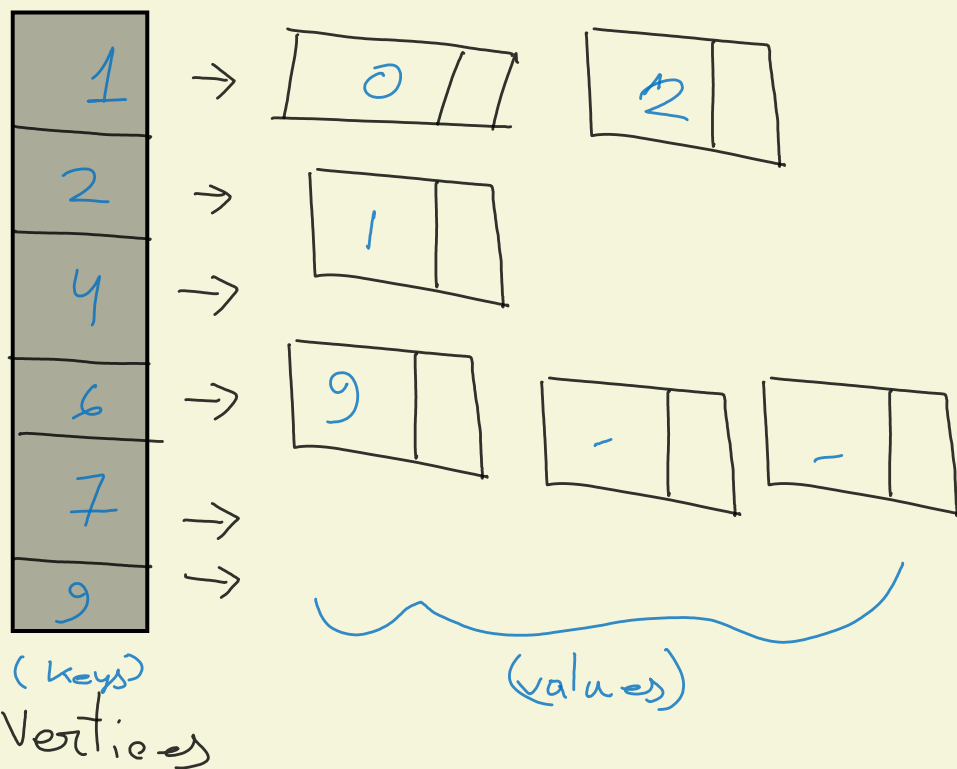
$$a[2][3] = 5$$

$$a[7][8] = 0$$

Concepts-

- Dense Graph
- Sparse Graph

Adjacency List Representation



e = a array of pointers, C++/Java = map, vector
Python = dictionary